

PIZZA SALES DATA ANALYSIS USING SQL

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B.TECH CSE | DATA ANALYTICS
PROJECT

Project Overview

This project analyzes pizza sales data using SQL to understand sales performance, customer ordering patterns, popular pizza types, and revenue generation.

The project uses SQL queries to extract meaningful insights from the pizza sales database and convert raw transactional data into useful business information.

Key Areas:

- Order analysis
- Revenue analysis
- Pizza popularity
- Customer ordering patterns
- Category-wise sales
- Time-based analysis
- Revenue contribution



Project Objectives

- Analyze the total number of orders.
- Calculate total revenue generated.
- Identify the most popular pizzas.
- Analyze pizza sizes and categories.
- Understand ordering patterns by time.
- Calculate average pizzas ordered per day.
- Identify top-performing pizzas based on revenue.
- Analyze the contribution of each pizza type to total revenue.
- Study cumulative revenue over time.

Dataset & Database

Tables :

1. Orders

- order_id
- order_date
- order_time

2. Order Details

- order_details_id
- order_id
- pizza_id
- quantity

3. Pizzas

- pizza_id
- pizza_type_id
- size
- price

4. Pizza Types

- pizza_type_id
- name
- category
- ingredients

Technologies Used

- SQL
- MySQL / SQL Server
- Excel / CSV Dataset
- SQL queries for data analysis



Retrieve the total number of orders placed.

Query

```
select count(order_id) as orders from orders;
```

Output

Result Grid	
	orders
▶	21350

Calculate the total revenue generated from pizza sales.

Query

```
SELECT  
    ROUND(SUM(order_details.quantity * pizzas.price),  
          2) AS total_revenue  
FROM  
    order_details  
    JOIN  
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Output

Result Grid	
	total_revenue
▶	817860.05

Identify the highest-priced pizza.

Query

```
SELECT  
    pizza_types.name, pizzas.price  
FROM  
    pizza_types  
    JOIN  
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id  
ORDER BY pizzas.price DESC limit 1;
```

Output

Result Grid			Filter Rows:
	name	price	
▶	The Greek Pizza	35.95	

Identify the most common pizza size ordered.

Query

```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
    JOIN
        order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC
LIMIT 1;
```

Output

Result Grid			Filter
	size	order_count	
▶	L	18526	

List the top 5 most ordered pizza types along with their quantities.

Query

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Output

Result Grid			Filter Rows:
	name	quantity	
▶	The Classic Deluxe Pizza	2453	
	The Barbecue Chicken Pizza	2432	
	The Hawaiian Pizza	2422	
	The Pepperoni Pizza	2418	
	The Thai Chicken Pizza	2371	

Join the necessary tables to find the total category of each pizza ordered.

Query

```
SELECT
    pizza_types.category, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC
LIMIT 5;
```

Output

Result Grid   		
	category	quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

Determine the distribution of orders by hour of the day.

Query

```
SELECT
    HOUR(order_time), COUNT(order_id)
FROM
    orders
GROUP BY HOUR(order_time);
```

Output

Result Grid   Filter Rows:		
	HOUR(order_time)	COUNT(order_id)
▶	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336
Result 9 x 		

Join relevant tables to find the category-wise distribution of pizzas. Group the orders by date and calculate the average number of pizzas ordered per day.

Query

```
select category ,count(name ) from pizza_types  
group by category;
```

Output

Result Grid			Filter R
	category	count(name)	
▶	Chicken	6	
	Classic	8	
	Supreme	9	
	Veggie	9	

Determine the top 3 most ordered pizza types based on revenue.

Query

```
SELECT pizza_types.name ,
sum(order_details.quantity* pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id =pizzas.pizza_type_id
join order_details
on pizzas.pizza_id = order_details.pizza_id
group by pizza_types.name
order by revenue desc limit 3;
```

Output

Result Grid				 Filter Rows:
	name	revenue		
▶	The Thai Chicken Pizza	43434.25		
	The Barbecue Chicken Pizza	42768		
	The California Chicken Pizza	41409.5		

Calculate the percentage contribution of each pizza type to total revenue.

Query

```
select pizza_types.category ,
round(sum(order_details.quantity* pizzas.price) /( select round(sum(order_details.quantity* pizzas.price),2) as total_

from order_details
join pizzas
on pizzas.pizza_id =order_details.pizza_id )* 100,2) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id =pizzas.pizza_type_id
join order_details
on pizzas.pizza_id = order_details.pizza_id
group by  pizza_types.category
order by revenue desc limit 3;
```

Output

Result Grid					Filter
	category	revenue			
▶	Classic	26.91			
	Supreme	25.46			
	Chicken	23.96			

**Thank
you!**